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EDUCATION

University of Toronto

Toronto, ON

Honours Bachelor of Science in Computer Science

Sep. 2020 – June 2026

- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Software Design, Operating Systems, Artificial Intelligence, Web Development, Algorithm Design & Analysis, Deep Learning & Neural Networks.
- **Activities:** 2021 and 2024 Dean's List, Computer Science Students' Union Council member, Executive member in the Powerlifting club and Webmaster in the Visual Arts Club.

EXPERIENCE

Machine Learning Research Assistant

May 2023 – April 2025

University of Toronto — CoNSens Laboratory

Toronto, ON

- Generated simulated visual cortex data in collaboration with the CoNSens Lab, directly supporting the findings of a cognitive neuroscience **research paper** supervised by **Professor Matthias Niemeier**.
- Achieved a **96% validation accuracy** in object recognition by engineering a custom **Convolutional Neural Network (CNN)** in **PyTorch** to simulate visual cortex processing in a research experiment.
- Optimized model architecture and performance by integrating advanced **attention modulation** and **fine-tuning** techniques sourced from cutting-edge computer vision literature.

Software Engineer Intern

June 2024 — August 2024

Wayfair Canada

Toronto, ON

- Streamlined the **machine learning training and fine-tuning pipeline** for the ML Platforms team, driving improved platform efficiency for **150+ clients**.
- Strengthened internal cloud infrastructure and database reliability by developing secure storage solutions and implementing real-time monitoring for ML models via **Google Cloud Platform (GCP)** and **Datadog**.
- Boosted database performance by **300%** and **reduced error rates by 50%** by identifying and resolving a critical concurrency bug in the ML database storage system.

Software Engineer Intern

September 2022 — December 2022

Microsoft (Nuance Communications)

Montreal, QC

- Optimized an internal employee web portal serving **2000+ users** by developing **React** and **JavaScript** features, partnering with UI/UX and QA teams to accelerate sprint delivery times.
- Directed a team of 3 developers to build a **React Native** autonomous shopping mobile application within a 48-hour Microsoft Hackathon, successfully prototyping a cashier-less checkout experience akin to Amazon Go.

Quality Engineer Intern

May 2022 — August 2022

Equifax Canada

Toronto, ON

- Ensured **100% data accuracy** during a critical user migration to **Google Cloud Platform (GCP)** by executing rigorous API validation testing via **Postman** and **Excel**.
- Identified and documented data synchronization defects across the full **Software Development Life Cycle (SDLC)**, accelerating issue resolution and preventing downstream database errors.

PROJECTS

Restify | *React.js, Django, Bootstrap, Python*

January 2023 – April 2023

- Developed a scalable web application with **React** and **Django**, simulating a complete Airbnb user flow from property discovery to final reservation and review processing.
- Integrated complex frontend UI components with custom **REST APIs**, seamlessly handling data exchange for account management and mock transactions, and dynamically rendering scalable property listing feeds.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, HTML/CSS, SQL, Java, C, C++, C#

Frameworks: Next.js, React.js, Django, Bootstrap, Node.js

Technologies: Git, MySQL, MongoDB, Firebase, Excel, Docker, Google Cloud Platform, Kubernetes, Visual Studio

Libraries: PyTorch, pandas, NumPy, Matplotlib